# EVOCURE Candidate Screening

Pancreatic ductal adenocarcinoma, five prioritised compounds

EVOCURE-PDAC-20261027 | generated 2026-10-27 21:46:37 UTC

**What this document is**

A demonstration of an open source in silico screening platform. It shows what the software computed on one reproducible run: a bounded evolutionary search over molecular structures, scored by a QSAR model, toxicology heuristics and a simulated two dimensional tumour microenvironment.

What it is not: evidence that any compound here treats pancreatic cancer. No molecule in this report has been synthesised, assayed in a cell, or given to an animal or a person. These are computational hypotheses at the earliest possible stage of discovery, and the large majority of such hypotheses fail.

## Result in one line

**No candidate of 5 separated from the negative control, and neither did the positive control when scored by the same potency model.**

The screening pipeline ran end to end and produced five ranked candidates. The tissue assay was checked with a positive and a negative control before the candidates were read, so the result above is interpretable rather than merely generated. The positive control separates from the negative control, so the tissue model responds to a known active in this configuration.

## Run at a glance

| | |
|---|---|
| Structures evaluated | 109 unique valid structures |
| Generations | 4, population 24 |
| Replicate seeds per compound | 5 |
| Compute backend | NumPy CPU float64 |
| QSAR training set | 43 curated compounds |
| QSAR scaffold split R2 | 0.323 |
| QSAR classifier ROC AUC | 0.917 |
| Wall clock | 322 seconds |

Open source platform and data. Reproduce this document with the commands in the reproducibility section. Report generated directly from the run's results file; no figure in it was entered by hand.

# 1. Summary

A bounded evolutionary search started from four approved pancreatic cancer drugs and generated structural variants through point mutations and scaffold recombination. Every variant was validated with RDKit, scored on five objectives, and the non dominated front was carried into the next generation. The five compounds below are the highest ranked variants that were not themselves seeds.

## 1.1 Assay validity, checked before reading the candidates

| Role | Compound | TGI % | 95% CI | CI excludes 0 |
|---|---|---|---|---|
| Positive, curated | Gemcitabine | 67.1 | 60.4 to 73.8 | yes |
| Negative | Caffeine | -5.7 | -18.3 to 6.8 | no |
| Positive, modelled | Gemcitabine | -1.0 | -14.7 to 12.7 | no |

The third arm is the same positive control molecule with its curated bioactivity withheld, so its potency comes from the structure-property model that scores every generated candidate. It is the only arm the candidates can fairly be compared against.

**Does the simulated assay work?**
Positive versus negative control: $p < 0.001$, mean difference 72.9 percentage points, Hedges g 8.12.

The positive control separates from the negative control, so the tissue model responds to a known active in this configuration.

## 1.1a The comparison is between a measurement and a prediction

**The positive control fails its own test when scored like a candidate**
Gemcitabine draws its potency from curated experimental bioactivity: IC50 0.0450 uM, and it inhibits growth by 67.1 %. A generated structure has no measurement, so it is scored by the structure-property model. Scoring this same molecule that way gives IC50 8.053 uM, 179 times weaker, and growth inhibition falls to -1.0 %, which does not separate from the negative control.

Nothing about the molecule changed. Only the provenance of its potency did. Any candidate in this report is therefore being compared against a number no generated structure can ever be assigned, and failing that comparison says little about the chemistry.

## 1.2 The five candidates

| ID | Compound | pIC50 | Score | TGI % | 95% CI | vs control |
|---|---|---|---|---|---|---|
| Candidate 01 | Compound-CFB9B6 | 7.03 | 0.572 | 9.4 | -6.8 to 25.5 | not distinguishable |
| Candidate 02 | Compound-5D822D | 7.13 | 0.564 | 5.5 | -6.8 to 17.9 | not distinguishable |
| Candidate 03 | Compound-FAFE93 | 5.97 | 0.544 | 1.6 | -8.9 to 12.2 | not distinguishable |
| Candidate 04 | Compound-C5F9E6 | 6.97 | 0.542 | 5.2 | -5.3 to 15.6 | not distinguishable |
| Candidate 05 | Compound-32914B | 7.10 | 0.540 | 0.5 | -13.5 to 14.4 | not distinguishable |

TGI is tumour growth inhibition in the simulated microenvironment, relative to an untreated arm, averaged over independent seeds. Score is the weighted composite across the five search objectives. The last column reports Welch's t-test of each candidate against the negative control; it is the column that decides whether a point estimate means anything.

**Read this before using the ranking**

No candidate separated from the negative control in the tissue simulation. The ranking above is therefore driven by predicted properties, drug likeness, toxicity alerts and applicability domain, not by a demonstrated simulated effect.

This is not evidence that the molecules are inactive. The positive control fails the identical test when scored by the same potency model, so the test does not currently distinguish a good candidate from a known drug. What the run establishes is a limit of the platform, not a property of these five structures.

The honest reading is that the search found chemically reasonable, low alert, drug like molecules near the seed scaffolds, and that this experiment cannot say whether any of them is active. Treat the list as a starting set for further computational work, not as a shortlist of likely treatments.

# 2. Method

## 2.1 Search

| | |
|---|---|
| Objective prompt | Find 5 experimental candidates for pancreatic cancer based on Gemcitabine, 5-Fluorouracil, Erlotinib and Paclitaxel. Prioritize activity, low predicted toxicity and drug-likeness. |
| Planner | local_rule_based_planner |
| Derived priorities | safety, druglikeness, activity |
| Seed compounds | Gemcitabine, 5-Fluorouracil, Paclitaxel, Erlotinib |
| Generations | 4 |
| Population size | 24 |
| Applied concentration | 5.0 uM |
| Exposure duration | 24.0 hours |
| Search seed | 20261027 |

The planner converts a plain language objective into search priorities only. It does not propose molecules and it does not supply any activity, potency or safety number. All quantitative output comes from the deterministic scientific models.

## 2.2 Objective weights

| Objective | Weight | Source of the value |
|---|---|---|
| Predicted PDAC activity | 2.0 | Calibrated QSAR regressor, sigmoidal utility at pIC50 6.0 |
| Low toxicological risk | 2.0 | Ten category structural alert and rule based risk index |
| Drug likeness (QED) | 2.0 | RDKit quantitative estimate of drug likeness |
| Selectivity index | 1.0 | ADMET derived tumour versus healthy cell window |
| Applicability domain | 1.0 | Maximum Tanimoto similarity to the training set |
| Simulated growth inhibition | 1.0 | Agent based tumour microenvironment, treated versus untreated |

Tumour cell kill is never maximised alone. A compound that scores well on activity but triggers toxicity alerts or falls outside the applicability domain is penalised by construction.

## 2.2a What selection actually optimises

**The selection loop is closed**

Every candidate that could be dosed was simulated in the tumour microenvironment, 178 runs in total, and its growth inhibition entered the objective set used for Pareto sorting and survivor selection. The tumour model therefore exerts selective pressure on the population rather than only reporting on the winners.

Two limits on how much that is worth. All arms share one seed per generation, so the comparison is paired but the objective is the noisiest of the six. And the tissue model's drug response is driven by the predicted IC50, so this objective is correlated with the activity objective rather than independent of it. It adds diffusion, uptake, hypoxia and clonal structure on top of that prediction; it does not replace it with a measurement.

## 2.3 The QSAR model, and what it can carry

| | |
|---|---|
| Training compounds | 43 |
| Validation | Bemis-Murcko scaffold split, no scaffold leakage |
| Regression RMSE | 1.704 log units |
| Regression R2 | 0.323 |
| Classifier ROC AUC | 0.917 |

| | |
|---|---|
| Classifier Brier score | 0.163 |
| Conformal 95% half width | 4.16 log units |

**The potency numbers in this report are weak, by the model's own measurement**

The pIC50 regressor explains little of the variance in the held out scaffolds (R2 = 0.32), and its calibrated 95% conformal interval is about 4.2 log units wide in each direction. An interval that wide spans roughly eight orders of magnitude in IC50, which is to say it does not constrain potency at all.

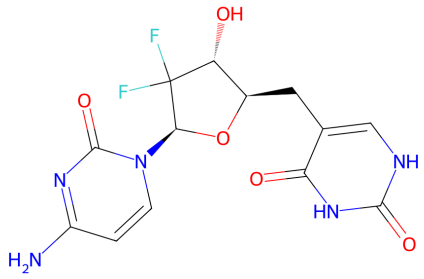
The active versus inactive classifier is the stronger half of the model (ROC AUC = 0.92). Reading pIC50 as a rough ordering signal is defensible. Reading any single predicted IC50 as a quantity is not. Every pIC50 in this report is printed with its conformal interval for that reason.

## 2.4 Tissue simulation

| | |
|---|---|
| Model | Two dimensional agent based PDAC microenvironment |
| Grid | 36 x 28 cells over 800 x 600 micrometres |
| Fields | Oxygen and nutrient reaction diffusion, drug diffusion and uptake |
| Features | Hypovascularity, dense collagen matrix, sensitive and resistant clones |
| Comparison | Treated arm against an untreated arm with the same seed |
| Replicates | 5 independent seeds per compound |

Kinetics and microenvironment parameters are exploratory and have not been fitted to experimental pancreatic tumour data. The model's drug response is driven by the predicted IC50, so two compounds with the same predicted IC50 produce the same trajectory. Simulated growth inhibition is a property of this model, not a measurement.

# Candidate 01: Compound-CFB9B6




## Identity

| | |
|---|---|
| Canonical SMILES | Nc1ccn([C@@H]2O[C@H](Cc3c[nH]c(=O)[nH]c3=O)[C@@H](O)C2(F)F)c(=O)n1 |
| InChIKey | SYMMKNWOSBOWTM-GTNGPMTGSA-N |
| Molecular formula | C13H13F2N5O5 |
| Murcko scaffold | O=c1[nH]cc(CC2CCC(n3cccnc3=O)O2)c(=O)[nH]1 |
| Origin | generation 4 variant, directed_optimization |

## Physicochemical profile

| MW | cLogP | TPSA | HBD | HBA | QED | Lipinski |
|---|---|---|---|---|---|---|
| 357.3 | -1.66 | 156.1 | 4 | 7 | 0.512 | passes |

## Predicted activity, with its uncertainty

| Quantity | Value | 95% interval | Reading |
|---|---|---|---|
| Predicted pIC50 | 7.030 | 2.87 to 11.19 | ordering signal only |
| Implied IC50 (uM) | 0.0933 | spans orders of magnitude | not a quantity |
| Active probability | 0.835 | 0.73 to 0.93 | the stronger prediction |
| Applicability domain | 0.64 | Tanimoto, nearest neighbour | inside domain |

## Objective scores and risk

| Activity | Selectivity | QED | Low tox | Domain | Composite | Risk index |
|---|---|---|---|---|---|---|
| 0.737 | 0.000 | 0.512 | 0.916 | 0.643 | 0.572 | 8.4 / 100 |

## Simulated tissue response

| Metric | Value |
|---|---|
| Tumour growth inhibition, mean | 9.38 % over 5 seeds |
| 95% confidence interval | -6.75 % to 25.52 % |
| Interval excludes zero | no |
| Per seed values | 17.1, -8.1, 26.2, 6.8, 4.9 |
| Resistant clone fraction | 0.345 |
| Value used as selection objective | 17.1 % on the search seed |
| Versus negative control | p = 0.076, Hedges g 1.17 |
| Distinguishable, uncorrected | no |
| Distinguishable, Bonferroni corrected | no |

## Relation to known drugs

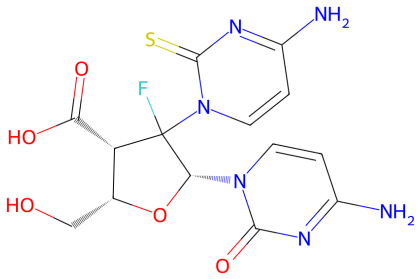
| | |
|---|---|
| Nearest reference | Gemcitabine (Tanimoto 0.643) |
| Exact structural match | none in the local library |
| Library scope | 28 curated reference drugs |

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

**Verdict**

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

# Candidate 02: Compound-5D822D




## Identity

| | |
|---|---|
| Canonical SMILES | Nc1ccn([C@@H]2O[C@H](CO)[C@H](C(=O)O)C2(F)n2ccc(N)nc2=S)c(=O)n1 |
| InChIKey | ZIEJAUJMRFDULL-SWVGTRJBSA-N |
| Molecular formula | C14H15FN6O5S |
| Murcko scaffold | O=c1ncccn1C1OCCC1n1cccnc1=S |
| Origin | generation 4 variant, directed_optimization |

## Physicochemical profile

| MW | cLogP | TPSA | HBD | HBA | QED | Lipinski |
|---|---|---|---|---|---|---|
| 398.4 | -0.75 | 171.5 | 4 | 9 | 0.480 | passes |

## Predicted activity, with its uncertainty

| Quantity | Value | 95% interval | Reading |
|---|---|---|---|
| Predicted pIC50 | 7.135 | 2.97 to 11.30 | ordering signal only |
| Implied IC50 (uM) | 0.0733 | spans orders of magnitude | not a quantity |
| Active probability | 0.838 | 0.74 to 0.94 | the stronger prediction |
| Applicability domain | 0.60 | Tanimoto, nearest neighbour | inside domain |

## Objective scores and risk

| Activity | Selectivity | QED | Low tox | Domain | Composite | Risk index |
|---|---|---|---|---|---|---|
| 0.757 | 0.000 | 0.480 | 0.916 | 0.597 | 0.564 | 8.4 / 100 |

## Simulated tissue response

| Metric | Value |
|---|---|
| Tumour growth inhibition, mean | 5.55 % over 5 seeds |
| 95% confidence interval | -6.77 % to 17.87 % |
| Interval excludes zero | no |
| Per seed values | 17.1, -8.1, 11.9, 6.8, 0.0 |
| Resistant clone fraction | 0.333 |
| Value used as selection objective | 17.1 % on the search seed |
| Versus negative control | p = 0.113, Hedges g 1.02 |
| Distinguishable, uncorrected | no |
| Distinguishable, Bonferroni corrected | no |

## Relation to known drugs

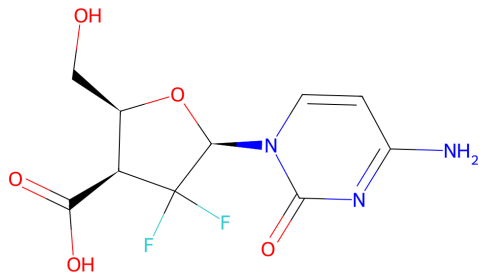
| | |
|---|---|
| Nearest reference | Gemcitabine (Tanimoto 0.597) |
| Exact structural match | none in the local library |
| Library scope | 28 curated reference drugs |

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

**Verdict**

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

# Candidate 03: Compound-FAFE93




## Identity

| | |
|---|---|
| Canonical SMILES | Nc1ccn([C@@H]2O[C@H](CO)[C@H](C(=O)O)C2(F)F)c(=O)n1 |
| InChIKey | HKQIZRDZFOYESW-UEYSZJFGSA-N |
| Molecular formula | C10H11F2N3O5 |
| Murcko scaffold | O=c1ncccn1C1CCCO1 |
| Origin | generation 4 variant, directed_optimization |

## Physicochemical profile

| MW | cLogP | TPSA | HBD | HBA | QED | Lipinski |
|---|---|---|---|---|---|---|
| 291.2 | -0.95 | 127.7 | 3 | 6 | 0.655 | passes |

## Predicted activity, with its uncertainty

| Quantity | Value | 95% interval | Reading |
|---|---|---|---|
| Predicted pIC50 | 5.971 | 1.81 to 10.13 | ordering signal only |
| Implied IC50 (uM) | 1.0685 | spans orders of magnitude | not a quantity |
| Active probability | 0.740 | 0.64 to 0.84 | the stronger prediction |
| Applicability domain | 0.77 | Tanimoto, nearest neighbour | inside domain |

## Objective scores and risk

| Activity | Selectivity | QED | Low tox | Domain | Composite | Risk index |
|---|---|---|---|---|---|---|
| 0.493 | 0.000 | 0.655 | 0.916 | 0.766 | 0.544 | 8.4 / 100 |

## Simulated tissue response

| Metric | Value |
|---|---|
| Tumour growth inhibition, mean | 1.63 % over 5 seeds |
| 95% confidence interval | -8.93 % to 12.19 % |
| Interval excludes zero | no |
| Per seed values | 0.0, -5.4, 9.5, 11.4, -7.3 |
| Resistant clone fraction | 0.312 |
| Value used as selection objective | 0.0 % on the search seed |
| Versus negative control | p = 0.248, Hedges g 0.71 |
| Distinguishable, uncorrected | no |
| Distinguishable, Bonferroni corrected | no |

## Relation to known drugs

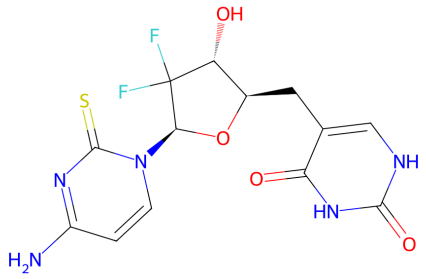
| | |
|---|---|
| Nearest reference | Gemcitabine (Tanimoto 0.766) |
| Exact structural match | none in the local library |
| Library scope | 28 curated reference drugs |

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

**Verdict**

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

# Candidate 04: Compound-C5F9E6




## Identity

| | |
|---|---|
| Canonical SMILES | Nc1ccn([C@@H]2O[C@H](Cc3c[nH]c(=O)[nH]c3=O)[C@@H](O)C2(F)F)c(=S)n1 |
| InChIKey | CONBZSYIZIVTIT-GTNGPMTGSA-N |
| Molecular formula | C13H13F2N5O4S |
| Murcko scaffold | O=c1[nH]cc(CC2CCC(n3cccnc3=S)O2)c(=O)[nH]1 |
| Origin | generation 4 variant, directed_optimization |

## Physicochemical profile

| MW | cLogP | TPSA | HBD | HBA | QED | Lipinski |
|---|---|---|---|---|---|---|
| 373.3 | -0.29 | 139.0 | 4 | 7 | 0.542 | passes |

## Predicted activity, with its uncertainty

| Quantity | Value | 95% interval | Reading |
|---|---|---|---|
| Predicted pIC50 | 6.967 | 2.80 to 11.13 | ordering signal only |
| Implied IC50 (uM) | 0.1079 | spans orders of magnitude | not a quantity |
| Active probability | 0.858 | 0.76 to 0.96 | the stronger prediction |
| Applicability domain | 0.52 | Tanimoto, nearest neighbour | inside domain |

## Objective scores and risk

| Activity | Selectivity | QED | Low tox | Domain | Composite | Risk index |
|---|---|---|---|---|---|---|
| 0.725 | 0.000 | 0.542 | 0.916 | 0.516 | 0.542 | 8.4 / 100 |

## Simulated tissue response

| Metric | Value |
|---|---|
| Tumour growth inhibition, mean | 5.16 % over 5 seeds |
| 95% confidence interval | -5.27 % to 15.58 % |
| Interval excludes zero | no |
| Per seed values | 0.0, -2.7, 19.1, 4.5, 4.9 |
| Resistant clone fraction | 0.299 |
| Value used as selection objective | 0.0 % on the search seed |
| Versus negative control | p = 0.102, Hedges g 1.06 |
| Distinguishable, uncorrected | no |
| Distinguishable, Bonferroni corrected | no |

## Relation to known drugs

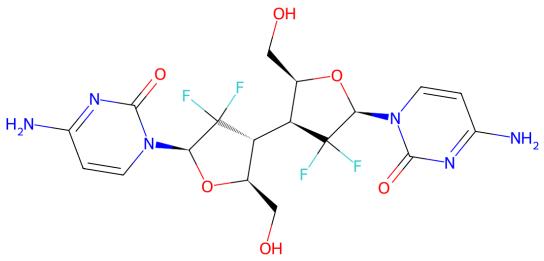
| | |
|---|---|
| Nearest reference | Gemcitabine (Tanimoto 0.516) |
| Exact structural match | none in the local library |
| Library scope | 28 curated reference drugs |

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

**Verdict**

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

# Candidate 05: Compound-32914B




## Identity

| | |
|---|---|
| Canonical SMILES | Nc1ccn([C@@H]2O[C@H](CO)[C@@H]([C@H]3[C@@H](CO)O[C@@H](n4ccc(N)nc4=O)C3(F)F)C2(F)F)c(=O)n1 |
| InChIKey | LXFJNEGRNGGPAR-YIDOWPQCSA-N |
| Molecular formula | C18H20F4N6O6 |
| Murcko scaffold | O=c1ncccn1C1CC(C2COC(n3cccnc3=O)C2)CO1 |
| Origin | generation 4 variant, directed_optimization |

## Physicochemical profile

| MW | cLogP | TPSA | HBD | HBA | QED | Lipinski |
|---|---|---|---|---|---|---|
| 492.4 | -1.05 | 180.7 | 4 | 10 | 0.376 | passes |

## Predicted activity, with its uncertainty

| Quantity | Value | 95% interval | Reading |
|---|---|---|---|
| Predicted pIC50 | 7.103 | 2.94 to 11.27 | ordering signal only |
| Implied IC50 (uM) | 0.0788 | spans orders of magnitude | not a quantity |
| Active probability | 0.810 | 0.71 to 0.91 | the stronger prediction |
| Applicability domain | 0.82 | Tanimoto, nearest neighbour | inside domain |

## Objective scores and risk

| Activity | Selectivity | QED | Low tox | Domain | Composite | Risk index |
|---|---|---|---|---|---|---|
| 0.751 | 0.000 | 0.376 | 0.896 | 0.818 | 0.540 | 10.4 / 100 |

## Simulated tissue response

| Metric | Value |
|---|---|
| Tumour growth inhibition, mean | 0.49 % over 5 seeds |
| 95% confidence interval | -13.47 % to 14.45 % |
| Interval excludes zero | no |
| Per seed values | 0.0, -13.5, 11.9, 11.4, -7.3 |
| Resistant clone fraction | 0.288 |
| Value used as selection objective | 0.0 % on the search seed |
| Versus negative control | p = 0.384, Hedges g 0.53 |
| Distinguishable, uncorrected | no |
| Distinguishable, Bonferroni corrected | no |

## Relation to known drugs

| | |
|---|---|
| Nearest reference | Gemcitabine (Tanimoto 0.818) |
| Exact structural match | none in the local library |
| Library scope | 28 curated reference drugs |

The reference library is small and local. Absence of a match here does not establish that a structure is novel, and similarity to an active drug does not establish that it is active. A patent and literature search would be the next step before treating any of these as new chemical matter.

**Verdict**

This candidate did not separate from the negative control. Its position in the ranking reflects predicted properties and the absence of alerts, not a demonstrated simulated effect.

# 3. What this run does not show

The value of a screening report lies as much in its boundaries as in its ranking. The following are not minor caveats; they determine what the numbers above can be used for.

**No experimental evidence of any kind**
No compound here has been synthesised. There is no cell assay, no animal model and no clinical data. Every number is the output of a model.

**The tissue model is not calibrated to real tumours**
Kinetics, diffusion constants, clone behaviour and drug response parameters are plausible but have not been fitted to experimental pancreatic tumour data. Simulated growth inhibition is a property of this software.

**The potency model cannot resolve potency**
A scaffold split R2 of 0.32 and a conformal interval of about 4.2 log units mean the regressor orders compounds loosely and quantifies none of them.

**A small, narrow training set**
43 curated compounds. Applicability domain is measured against that set, so high domain confidence means close to the training data, which also means structurally unadventurous.

**Virtual time is not real time**
Generations and exposure hours index the simulation. They do not correspond to chemical synthesis, biological timescales or clinical development.

**Structural similarity is not efficacy and not novelty**
Matching a known drug does not transfer its activity. Failing to match anything in a small local library does not make a structure new.

**No ADME, formulation or synthesis assessment**
Nothing here addresses whether these molecules can be made at reasonable cost, are stable, are orally available, or reach a pancreatic tumour in a living system. Dense desmoplastic stroma makes that last point a central problem in this disease.

**Multiple comparisons**
Five candidates were tested against one control arm. The per candidate threshold is uncorrected; a Bonferroni corrected threshold would be $p < 0.01$, and the results file records both.

**Reference drugs and novel structures are not scored the same way**
Gemcitabine's potency in the tissue model comes from curated experimental bioactivity. A newly generated structure has no measurement, so its potency comes from the structure-property model, which assigns micromolar values far weaker than the positive control's. At the dose used here that difference alone is close to sufficient to explain why no candidate matches the control. The comparison is therefore between a measurement and a prediction, not between two compounds, and failing to beat the control is weak evidence about the molecules.

**The tissue objective is noisy and not independent**
All arms in a generation share one seed, so selection sees a single draw of a quantity whose spread across seeds crosses zero. The objective also depends on the potency the pharmacology model assigns, so it is correlated with the activity objective rather than an independent check on it.

**If you take one thing from this report**
Pancreatic ductal adenocarcinoma has a five year survival rate in the low teens, and that is precisely why early computational output about it should be described carefully. This platform is a hypothesis generator. The appropriate next step for any compound listed here is scrutiny by a medicinal chemist, not optimism.

# 4. Reproducibility

## 4.1 Environment of this run

| | |
|---|---|
| Platform | Windows-11-10.0.26200-SP0 |
| Python | 3.12.13 |
| Compute backend | NumPy CPU float64 |
| Device | host CPU |
| Field precision | float64 |
| Search seed | 20261027 |
| Replicate seeds | 20261027, 20261028, 20261029, 20261030, 20261031 |
| Results hash | c2504b0fa0328c092e356ab2ca5b752c025eafaacfe0270c |

## 4.2 Library versions

| Package | Version |
|---|---|
| numpy | 2.2.6 |
| rdkit | 2026.9.1 |
| scikit-learn | 1.9.1 |
| scipy | 1.18.1 |
| torch | 2.11.0+cu128 |

## 4.3 Source file hashes, SHA-256

| File | SHA-256 (first 40 characters) |
|---|---|
| data\reference\pdac_reference_dataset.json | 10de36cbf66cb3f74e2f79dd40a355847247b3bd |
| models\scientific\pdac_qsar_v1.joblib | 088c53985fe5ece80dda27fb6a073c219292c1d5 |
| simulation\biology\diseases\pdac.py | 38d708f39a7977c0d8dadd07e9c72bffe219306f |
| simulation\core\engine.py | 4059a6fdea8603c90d4abe897475d3b48aaed4b0 |
| simulation\discovery\pipeline.py | 7aacc0fe78a202e3abcf096abed220e1ec83d0d0 |
| simulation\discovery\workbench.py | a1bf63f8e409aa7e407d316beee8c21a06a20fcb |
| simulation\pharmacology\admet.py | 064fe566b36565c07af097cb99a379543c6ddd4b |

## 4.4 Reproducing this document

The platform is open source and so is this result. The reference dataset, the pipeline, the tissue models and the report generator are all in the repository. No model weights are required: the QSAR artifact is rebuilt from the committed dataset, and the optional language models were not used for any quantitative output.

```
python scripts/train_qsar.py

# Linux and macOS
EVOCURE_COMPUTE=cpu EVOCURE_DISABLE_LAYA_NEURAL=1 \
  python scripts/run_pdac_report_experiment.py

# Windows PowerShell
$env:EVOCURE_COMPUTE = "cpu"
$env:EVOCURE_DISABLE_LAYA_NEURAL = "1"
python scripts\run_pdac_report_experiment.py

python scripts/build_candidate_pdf.py
```

Forcing the CPU backend is what makes a run bit reproducible. GPU backends that support float64 agree to within 1e-12 on the substrate fields; float32 backends such as DirectML and Metal differ at around 1e-6, which is harmless for the conclusions and fatal for hash

equality. The backend used is recorded above and in the results file.

### 4.5 Licence and reuse

The platform is released under the MIT licence and the curated reference dataset under CC-BY 4.0. Results, including this report and the underlying JSON, are published so that the discoveries stay open: anyone may reproduce, check, contradict or build on them without permission.